

Supplementary Material

Training and Validation Loss Curves

This supplementary material provides additional insights into the training dynamics of the proposed model. Figure 1 illustrates the evolution of the training and validation loss across the five folds used in the cross-validation procedure.

Across all folds, the model exhibits stable convergence behavior, with both training and validation losses decreasing smoothly over epochs. The validation curves closely follow the training curves, indicating consistent generalization and the absence of severe overfitting. Minor variations between folds are observed, which are expected due to differences in data composition across the splits. Overall, these results confirm the robustness of the optimization process and support the reliability of the reported cross-validation performance.

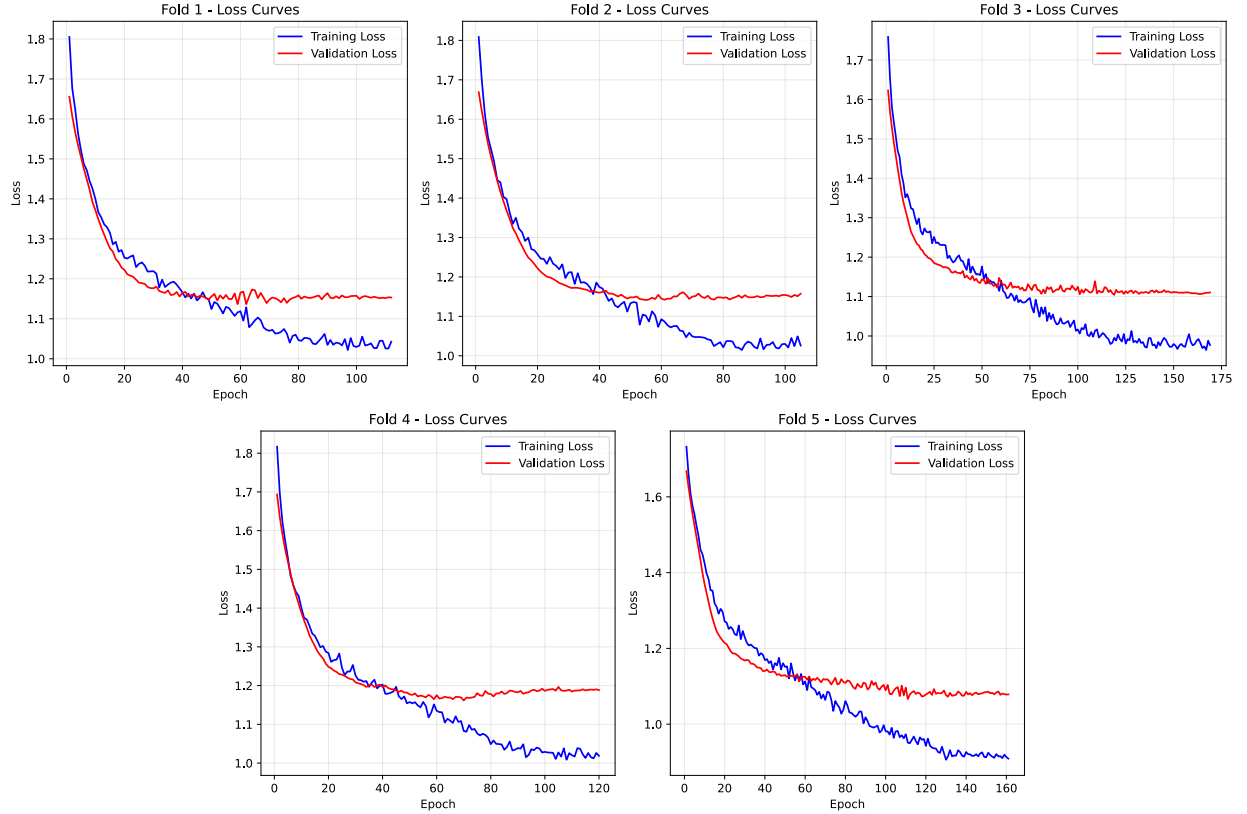


Figure 1: Training and validation loss curves for each fold of the five-fold cross-validation. The figure shows consistent convergence behavior across folds, with stable learning dynamics and no evident overfitting.